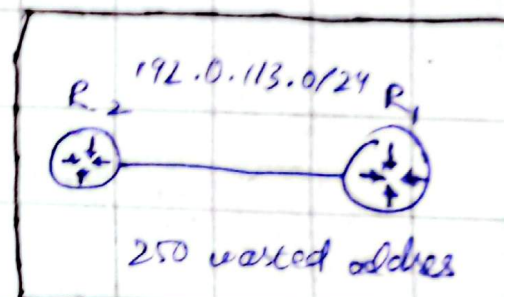


SUBNETTING

If we want to assign ip addresses to a network we have to follow IPv4 classes but they wasted many address

Example: If two routers R_1 and R_2 connect two each they need ~~4~~ addresses following address:

- 1- network address
- 2- Broadcast address
- 3- R_1 address
- 4- R_2 address



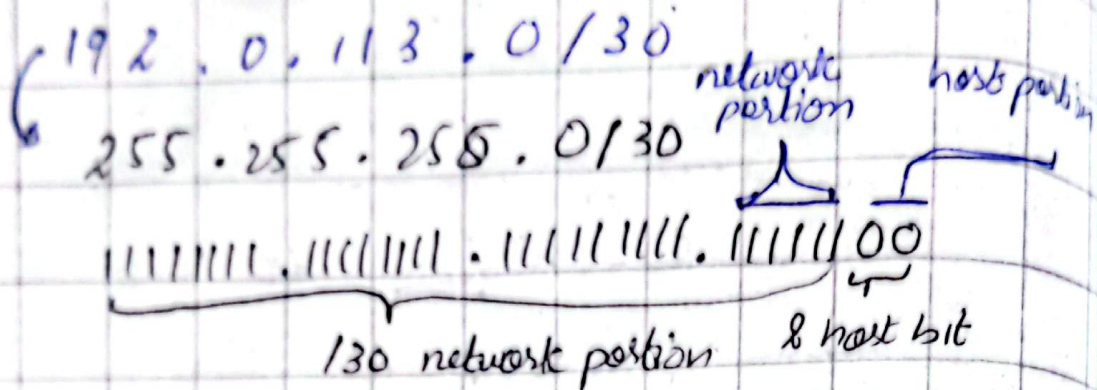
To calculate usable address we use $2^n - 2$ - n is host bit
network $\left\{ \begin{array}{l} \text{broadcast} \\ \text{cast} \end{array} \right\}$

But if we use class C (which smallest prefix) address will be wasted

($4 - 254 = 250$ wasted address)

CIDR (Classless Inter Domain Routing)
we use CIDR for above problem. It provide all prefixes from $18.11.16/24$ to $17.11.132$ etc

for 4 address we can use /30 why?



$$2^2 - 2 = 2 \text{ usable address (this include network + network and broadcast)}$$

now we can use /30 and no address is wasted but when a point to point connection we can also use only two address network and broadcast so we can use also /31

If we want 5 subnets in we will use

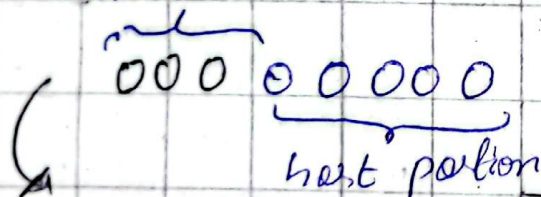
subnet formula for $2^1 = 2$ ✗

$$2^2 = 4 \text{ ✗}$$

$$2^3 = 8 \text{ ✓}$$

borrowing bit

→ correct we need 5 subnet



this mean we use /27 subnet

00000000
11110001

Subnetting in class A, B are also same

• For exact number of subnets we can 2^x where x is number of network bit. If we borrow 3 bit from host we get 8 subnet

• For usable address $2^n - 2 = \text{usable address}$

Class B address:

172.16.0.0/16

→ create 80 subnets

↳ $2^7 = 128$ subnets

0/16
10101101.00010000.00000000.00000000
7 bits

0/27 will be used for 80 subnets

- 172.22.0.0/16 divide this into 500 subnets and what prefix must be used

0/16
2⁹ = 512
10101101.00010110.00000000.00000000
9 bits

0/25 will be used for 500 subnets

Class A address:

PC1 has IP of 10.217.182.223/11

find following

- Network address 10.223.255.254
- last usable address
- broadcast address 10.223.255.255/11
- Number of usable addresses 2097152

00001010.11011001